The universal positive-block norm inequality forces an essentially Hermitian off-diagonal block

Principia Math in Progress

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Task 1. Resolve the essentially Hermitian conjecture

All matrices are complex and all norms below are operator norms. A matrix is *essentially Hermitian* if it has the form $\alpha I + \beta T$ with $T = T^*$ and $\alpha, \beta \in \mathbb{C}$. Equivalently, its numerical range is a line segment, possibly a point. We prove Conjecture 3.3 of Bourin–Lee [1], also Conjecture 7.3.3 on page 114 of Bourin's thesis [2].

Theorem 1. *Suppose $X \in M_n(\mathbb{C})$ satisfies*

$$\left\| \begin{pmatrix} A & X \\ X^* & B \end{pmatrix} \right\| \le \|A + B\| \quad \text{whenever } \begin{pmatrix} A & X \\ X^* & B \end{pmatrix} \ge 0. \tag{1}$$

Then X is essentially Hermitian. No invertibility or multiplicity restriction on X is necessary.

Proof. We first prove normality. For any $A > 0$, the matrix

$$M_A = \begin{pmatrix} A & X \\ X^* & X^*A^{-1}X \end{pmatrix} = \begin{pmatrix} A^{1/2} \\ X^*A^{-1/2} \end{pmatrix} \begin{pmatrix} A^{1/2} & A^{-1/2}X \end{pmatrix}$$

is positive semidefinite. Its nonzero eigenvalues are those of $A + A^{-1/2}XX^*A^{-1/2}$. Thus (1) implies

$$\left\| A + A^{-1/2}XX^*A^{-1/2} \right\| \le \left\| A + X^*A^{-1}X \right\|. \tag{2}$$

Fix a unit vector v and $0 < \varepsilon < 1$, and put $P = vv^*$ and $H = P + (1-\varepsilon)(I - P)$. For $n > 1$, H has simple largest eigenvalue 1 with eigenvector v; for $n = 1$ normality is automatic. Apply (2) with $A = tH$ and divide by t. With $u = t^{-2}$,

$$\lambda_{\max}(H + uH^{-1/2}XX^*H^{-1/2}) \le \lambda_{\max}(H + uX^*H^{-1}X).$$

For a Hermitian C, the simple-eigenvalue first variation is

$$\lim_{u \downarrow 0} \frac{\lambda_{\max}(H + uC) - 1}{u} = v^*Cv. \tag{3}$$

For completeness, a unit maximizing vector w_u converges to the line $\mathbb{C}v$ by the spectral gap of H. The Rayleigh principle bounds the quotient in (3) below by v^*Cv and above by $w_u^*Cw_u$, proving the formula. Consequently

$$\|X^*v\|^2 \le v^*X^*H^{-1}Xv.$$

Letting $\varepsilon \downarrow 0$ gives $v^*XX^*v \le v^*X^*Xv$ for every unit vector v. The positive semidefinite matrix $X^*X - XX^*$ has trace zero, so it is zero. Hence X is normal.

We next record a necessary symmetry of extreme eigenvalues. For any Hermitian H, define

$$K_X(H) = \begin{pmatrix} H & X \\ X^* & -H \end{pmatrix}, \qquad a = \lambda_{\max}(K_X(H)), \quad b = -\lambda_{\min}(K_X(H)).$$

The trace is zero, so $a, b \ge 0$. The matrix $bI_{2n} + K_X(H)$ is positive semidefinite and its two diagonal blocks sum to $2bI_n$. Condition (1) therefore gives $b + a \le 2b$, or $a \le b$. Writing $J = \operatorname{diag}(I_n, -I_n)$, we have $K_X(-H) = -JK_X(H)J$. Applying the same argument to $-H$ gives $b \le a$. Thus

$$\lambda_{\max}(K_X(H)) = -\lambda_{\min}(K_X(H)) \qquad (H = H^*). \tag{4}$$

By simultaneous unitary conjugation in (1), we may now assume X diagonal. Suppose three of its eigenvalues z_1, z_2, z_3 are noncollinear. Set

$$\Delta = \Im\big(\overline{z_2 - z_1}(z_3 - z_1)\big) \neq 0, \qquad D = \operatorname{diag}(z_1, z_2, z_3), \qquad L = \begin{pmatrix} 0 & i & 1 \\ -i & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Choose $h > 2\|X\|/\sqrt{3}$ and take $H = hL \oplus 0_{n-3}$. Up to a permutation, $K_X(H)$ is the direct sum of

$$K = \begin{pmatrix} hL & D \\ D^* & -hL \end{pmatrix}$$

and matrices with eigenvalues $\pm|z_j|$, $j > 3$. The eigenvalues of L are $0, \sqrt{3}, -\sqrt{3}$. The off-diagonal perturbation of $\operatorname{diag}(hL, -hL)$ has norm $\|D\| \le \|X\|$. The Rayleigh principle therefore gives

$$\lambda_{\max}(K) \ge \sqrt{3}h - \|X\| > \|X\|, \qquad \lambda_{\min}(K) \le -\sqrt{3}h + \|X\| < -\|X\|.$$

Both extreme eigenvalues of the full matrix consequently come from K. By (4) they are R and $-R$ for some $R > 0$.

We show that K cannot have any pair of opposite nonzero eigenvalues. Direct trace identities give

$$\operatorname{Tr} K = \operatorname{Tr} K^3 = 0, \qquad \operatorname{Tr} K^5 = -20h^3\Delta. \tag{5}$$

Here is an explicit verification of the only nontrivial identity. Write $K = S + T$, with $S = \operatorname{diag}(hL, -hL)$ and $T = \left(\begin{smallmatrix} 0 & D \\ D^* & 0 \end{smallmatrix}\right)$. Words with an odd number of T's have trace zero. Since $DD^* = D^*D$, the remaining contributions to $\operatorname{Tr} K^5$ reduce, by cyclicity, to

$$5h^3 \operatorname{Tr}\big(L^2(D^*LD - DLD^*)\big).$$

Indeed the two cyclic classes with two T's each have five words; $\operatorname{Tr}(S^3T^2) = 0$, and the class S^2TST is the expression displayed. The terms with zero or four T's cancel between the diagonal blocks. Using

$$L^2 = \begin{pmatrix} 2 & 1 & i \\ 1 & 2 & -i \\ -i & i & 2 \end{pmatrix},$$

the remaining trace equals

$$4\Im(z_1\overline{z}_2 - z_1\overline{z}_3 + z_2\overline{z}_3) = -4\Delta.$$

The cubic trace in (5) follows in the same way from $\operatorname{Tr} S^3 = 0$ and $\operatorname{Tr}(ST^2) = 0$.

Newton's identities now show that the characteristic polynomial has form

$$p(\lambda) = \lambda^6 + c_2\lambda^4 + c_4\lambda^2 + 4h^3\Delta\lambda + c_6.$$

In detail, the coefficients of λ^5 and λ^3 vanish because the first and third power traces vanish, and the coefficient of λ is $-\operatorname{Tr}(K^5)/5 = 4h^3\Delta$. But $p(R) = p(-R) = 0$ would give $8h^3\Delta R = 0$, a contradiction. Thus no three eigenvalues of X are noncollinear. The spectrum lies on a line (the assertion is also immediate for at most two distinct eigenvalues). Normality then gives $X = \alpha I + \beta T$ with $T = T^*$, as required. □

If "positive" is interpreted as positive definite, the same conclusion holds: add δI to both diagonal blocks of each semidefinite test and let $\delta \downarrow 0$. The source's known width-zero inequality gives the converse for essentially Hermitian X [1, 2].

Task 2. Quantify stability. For $\|X\| \le 1$, relate the distance from X to the essentially Hermitian matrices to the largest violation of (1) over positive test matrices with bounded norm, tracking dependence on n.

Task 3. Classify the Schatten exponents. For each $1 < p < \infty$, characterize the matrices X for which the universal inequality holds with the Schatten p-norm; compare the known normality characterization at $p = 2$ with the operator-norm result here.

Task 4. Minimize the detecting test families. Determine which restrictions on the ranks of the diagonal blocks and the full positive test matrix still suffice to detect every matrix that is not essentially Hermitian.

Task 5. Examine infinite-dimensional operators. Determine the corresponding universal-inequality class for bounded operators on an infinite-dimensional Hilbert space, where the finite-dimensional trace argument forcing normality is unavailable.

References

[1] J.-C. Bourin and E.-Y. Lee, *Eigenvalue inequalities for positive block matrices with the inradius of the numerical range*, International Journal of Mathematics 33 (2022), 2250009; arXiv:2111.15180, Conjecture 3.3, p. 6.

[2] J.-C. Bourin, *A Journey into Matrix Analysis*, arXiv:2307.03064 (2023), Conjecture 7.3.3, p. 114.